

ARCHITECTURAL RECOMMENDATION DOCUMENT

To: Executive Leadership Team, BSAVA

From: Timberyard

Date: July 7, 2026

Subject: Customer Identity & Access Management (CIAM) Selection for MACH Platform

Status: Under Review

1. Executive Summary

Following the strategic decision to decouple Single Sign-On (SSO) from Salesforce CRM, BSAVA requires a dedicated, modern Customer Identity and Access Management (CIAM) solution. The identity layer must seamlessly integrate with the new headless MACH ecosystem (Next.js, Contentful, Pimcore, Commerce Layer) while maintaining Salesforce as the Single Source of Truth (SSOT) for member data.

This document evaluates the two primary candidates: **Okta Customer Identity Cloud (powered by Auth0)** and **Microsoft Entra External ID**.

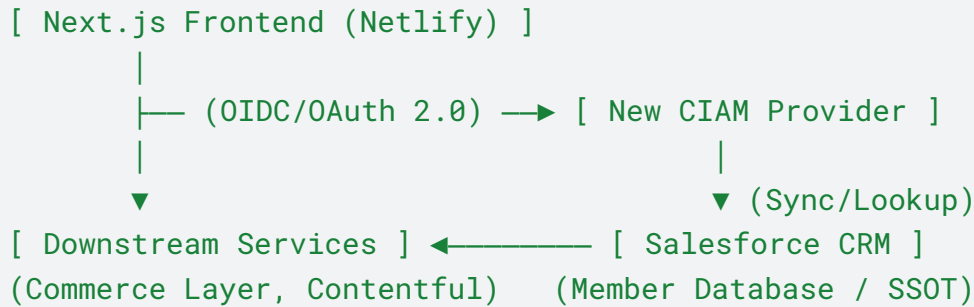
High-Level Recommendation: * If **total cost of ownership (TCO)** and **budget optimization** are the primary drivers, **Microsoft Entra External ID** is recommended due to its disruptive 50,000 free Monthly Active User (MAU) tier.

- If **developer velocity, rapid time-to-market, and frictionless integration** with the Next.js/Netlify stack are the priority, **Okta (Auth0)** is recommended as the technically superior fit for a modern MACH architecture.

2. Context & Architectural Drivers

The BSAVA platform is transitioning to a headless, composable architecture. The presentation layer (Next.js hosted on Netlify) requires an identity provider that operates via standard OpenID Connect (OIDC) and OAuth 2.0 protocols, issuing JSON Web Tokens (JWTs) to authorize downstream services at the edge.

None



The selected CIAM solution must address three critical architectural drivers:

1. **Performance at the Edge:** Fast token issuance and verification to minimize latency during member browsing, searching (Algolia), and checkout (Commerce Layer).
2. **Salesforce State Synchronization:** The ability to dynamically query or sync with Salesforce to append membership status into the JWT claims for real-time entitlement validation.
3. **Omnichannel User Experience:** A seamless, securely branded login flow for veterinarians accessing courses, events, books, and memberships.

3. Option Evaluation

Option A: Okta Customer Identity Cloud (Auth0)

Auth0 is a developer-centric, cloud-native CIAM platform designed specifically for modern web applications and decoupled architectures.

- **Architectural Fit:** Exceptional. The `@auth0/nextjs-auth0` SDK features mature, native support for the Next.js App Router and runs efficiently within Netlify's Edge Functions.
- **Extensibility (Auth0 Actions):** Auth0 allows developers to write custom Node.js serverless functions directly inside the authentication pipeline. When a user logs in, an Action can instantly query the Salesforce API, fetch the user's current membership tier, and inject it as a custom claim into the JWT. Downstream systems (like Contentful for gated content or Commerce Layer for member pricing) can read this token instantly without hitting Salesforce again.
- **User Experience:** Highly customizable template engine supporting pixel-perfect brand alignment, multi-branded flows, and sophisticated account-linking features (e.g., merging duplicate registrations seamlessly).
- **Risk/Con:** Premium pricing model. The free tier caps early, and transitioning to advanced enterprise features (custom domains, advanced MFA, higher SLAs) incurs significant subscription scaling costs.

Option B: Microsoft Entra External ID

Entra External ID is Microsoft's next-generation CIAM platform, built from the ground up to replace the legacy Azure AD B2C with a modern, developer-friendly interface.

- **Architectural Fit:** Strong. It relies on standard OAuth/OIDC protocols and utilizes the Microsoft Authentication Library (MSAL). It integrates cleanly into Next.js but lacks the tailored, edge-optimized developer wrappers found in the Auth0 ecosystem.
- **Extensibility:** Customization relies on "Custom Authentication Extensions," which trigger external REST APIs during the login flow. To fetch real-time membership data from Salesforce, BSAVA would need to deploy and manage intermediary infrastructure, such as Azure Functions or Netlify API routes, to orchestrate the API handshake.
- **User Experience:** Greatly improved over older Microsoft identity products. It offers clean, native, customizable branding templates and robust self-service flows, though it remains slightly less flexible than Auth0 for highly bespoke frontend alterations.
- **Commercial Fit (The Major Advantage):** Highly disruptive pricing. Microsoft provides the **first 50,000 Monthly Active Users (MAU) completely free**, with very competitive, flat utility pricing thereafter. For an association like BSAVA with a cyclical or large user base, this represents a massive operational cost saving.

4. Comparison Matrix

Criteria	Okta (Auth0)	Microsoft Entra External ID
Core Architecture	Cloud-native, pure-play CIAM	Enterprise-grade, tenant-isolated CIAM
Next.js / Netlify Integration	Native, edge-optimized SDKs; minimal friction	Standard MSAL integration; requires standard setup
Salesforce Integration	High (via embedded serverless Auth0 Actions)	Medium (requires external Azure Functions/webhooks)
Customization & UX	Fully flexible, developer-controlled templates	Template-driven, clean but more rigid
Pricing & TCO	High / Premium scaling	Very Low (First 50k MAUs free)
Operational Overhead	Low (Fully managed within identity pipeline)	Medium (Requires managing external middleware)

5. Definitive Recommendations & Next Steps

Recommendation Path A: Choose Microsoft Entra External ID

Trigger: *If minimizing operational software spend and leveraging existing corporate IT alignment are paramount.* If BSAVA's internal operational IT is already aligned with the Microsoft

365/Azure ecosystem, and the organization wants to keep fixed recurring costs to an absolute minimum, Entra External ID is the clear choice. The 50k free MAU allotment essentially commoditizes the identity layer, protecting BSAVA from scaling penalties as member engagement grows.

- **Architectural Action Item:** Provision an Entra External ID tenant and task the development team with building a proof-of-concept Custom Authentication Extension utilizing an API route to mock the Salesforce membership lookup.

Recommendation Path B: Choose Okta (Auth0)

Trigger: *If engineering speed, minimal infrastructure footprint, and maximum flexibility are the primary drivers.* If the priority is giving the development team the absolute best tool for a headless MACH stack to accelerate delivery of the new site, Auth0 is the industry leader. It eliminates the need to build external middleware to connect identity to Salesforce, handling the entire orchestration cleanly inside the identity platform via Auth0 Actions.

- **Architectural Action Item:** Request a formal volume quote from Okta tailored to BSAVA's projected MAU volume to review the exact financial delta against Entra before committing to a final purchase order.